WORKED SOLUTION

STEP III 2002, Question 3

Pure • Curve sketching

STEP III 2002, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q3

The key idea

Everything follows from the shape of the graph: it starts at $x = b$ with a vertical tangent going downwards, falls to a single minimum, and then rises without bound. Once we know the height of the starting point and the height of the minimum, counting solutions of $\mathrm{f}(x) = c$ is just a matter of drawing a horizontal line at height c and seeing how many times it meets the curve. The two numerical equations are then special cases whose values of c^2 land on the boundaries and in the interior of these ranges.

Solution

The graph

At the left-hand end, $\mathrm{f}(b) = a\sqrt{b}$. Differentiating, for $x > b$,

$$\mathrm{f}'(x) = \frac{a}{2\sqrt{x}} - \frac{1}{2\sqrt{x-b}}.$$

As $x \to b^+$ the second term tends to $-\infty$, so the curve leaves the point $(b, a\sqrt{b})$ heading straight down (a vertical tangent).

Since both square roots are positive,

$$\mathrm{f}'(x) < 0 \iff a\sqrt{x-b} < \sqrt{x} \iff a^2(x-b) < x \iff x < \frac{a^2 b}{a^2-1},$$

where the last step uses $a^2 - 1 > 0$. So f is strictly decreasing on $[b, x_0]$ and strictly increasing on $[x_0, \infty)$, where

$$x_0 = \frac{a^2 b}{a^2 - 1}.$$

At the minimum, $x_0 - b = \dfrac{b}{a^2-1}$, so

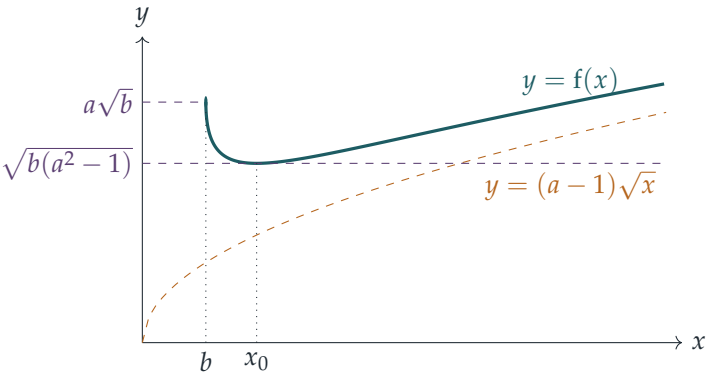
$$\mathrm{f}(x_0) = a \cdot \frac{a\sqrt{b}}{\sqrt{a^2-1}} - \frac{\sqrt{b}}{\sqrt{a^2-1}} = \frac{(a^2-1)\sqrt{b}}{\sqrt{a^2-1}} = \sqrt{b(a^2-1)}.$$

For large x, rationalising gives

$$\mathrm{f}(x) = (a-1)\sqrt{x} + (\sqrt{x} - \sqrt{x-b}) = (a-1)\sqrt{x} + \frac{b}{\sqrt{x} + \sqrt{x-b}},$$

so, because $a > 1$, $\mathrm{f}(x) \to \infty$, and the curve lies just above $y = (a-1)\sqrt{x}$, approaching it as $x \to \infty$.

Finally, the starting height is above the minimum: $a^2 b > (a^2-1)b$, so $a\sqrt{b} > \sqrt{b(a^2-1)}$.



Counting solutions

Write $m = \sqrt{b(a^2-1)}$ for the minimum value and $M = a\sqrt{b}$ for the starting value, so $0 < m < M$. On each of the two monotonic pieces, f is continuous and strictly monotonic, so it takes each value in its range exactly once:

- on $[b, x_0]$, f takes each value in $[m, M]$ exactly once;
- on $[x_0, \infty)$, f takes each value in $[m, \infty)$ exactly once.

The two pieces share only the point x_0, where the value is m.

Since $\mathrm{f}(x) \geq m$ for every $x \geq b$, the equation $\mathrm{f}(x) = c$ has no solution when $c < m$. For $c > 0$ this is the same as $c^2 < m^2$, that is,

$$c^2 < b(a^2-1) \quad \Longrightarrow \quad \text{no solution},$$

which is what we had to show. For the other cases:

? **Ask yourself** Why does $c^2 = a^2 b$ give two solutions rather than one?

At $c = M$ the left piece contributes the endpoint $x = b$ itself, which is a perfectly good solution because the domain includes $x = b$; the right piece contributes another one further out. Only when $c > M$ does the left piece stop contributing. So, for $c > 0$:

- **exactly one solution** when $c^2 = b(a^2-1)$ (just $x = x_0$, where the line touches the minimum) or when $c^2 > a^2 b$ (only the rising branch reaches that high);

- **exactly two solutions** when $b(a^2-1) < c^2 \leq a^2 b$ (one on each branch).

Part (i)

Here $a = 3$, $b = 2$, $c = 4$, so $b(a^2-1) = 16 = c^2$: the line touches the minimum, and there is exactly one solution,

$$x = x_0 = \frac{9 \cdot 2}{8} = \frac{9}{4}.$$

Indeed $3 \cdot \frac{3}{2} - \sqrt{\frac{1}{4}} = \frac{9}{2} - \frac{1}{2} = 4$. Directly: rearranging to $3\sqrt{x} - 4 = \sqrt{x-2}$ and squaring gives $8x - 24\sqrt{x} + 18 = 0$, that is, $(2\sqrt{x} - 3)^2 = 0$. The repeated root is the algebraic sign of tangency.

Part (ii)

Here $a = 3$, $b = 3$, $c = 5$, so $b(a^2-1) = 24 < 25 = c^2 < 27 = a^2 b$, and there are exactly two solutions. To find them, write the equation as $3\sqrt{x} - 5 = \sqrt{x-3}$ and square:

$$9x - 30\sqrt{x} + 25 = x - 3 \quad \Longrightarrow \quad 4x - 15\sqrt{x} + 14 = 0.$$

As a quadratic in $\sqrt{x}$, $(\sqrt{x} - 2)(4\sqrt{x} - 7) = 0$, so $\sqrt{x} = 2$ or $\sqrt{x} = \frac{7}{4}$, giving

$$x = 4 \qquad \text{or} \qquad x = \frac{49}{16}.$$

Squaring can introduce spurious roots, so we check both in the original equation. Both satisfy $x \geq 3$ (as $\frac{49}{16} = 3.0625$), and

$$3 \cdot 2 - \sqrt{1} = 5, \qquad 3 \cdot \tfrac{7}{4} - \sqrt{\tfrac{1}{16}} = \tfrac{21}{4} - \tfrac{1}{4} = 5.$$

So both are genuine, matching the count of two. As expected, one lies on each side of $x_0 = \frac{27}{8} = 3.375$.

Checks

- With $a = \frac{3}{2}$, $b = 2$ (the sketch), $x_0 = \frac{18}{5}$ and $f(\frac{18}{5}) = \frac{3}{2}\sqrt{3.6} - \sqrt{1.6} = \sqrt{2.5} \approx 1.581$, below $f(2) = \frac{3}{2}\sqrt{2} \approx 2.121$.
- A numerical scan of f over a fine grid for 200 random pairs (a, b) confirmed the minimum value $\sqrt{b(a^2-1)}$ at x_0, and counting crossings of $y = c$ gave 0, 2 and 1 solutions in the three ranges of c^2.
- In (i) the double root is consistent with c^2 sitting exactly at $b(a^2-1)$; in (ii) $c^2 = 25$ is strictly between 24 and 27, and we did find two roots.

Common slips

- Putting $c^2 = a^2 b$ in the "one solution" case: $x = b$ is in the domain, so it counts.

- Squaring without checking afterwards. Here both roots survive, but you must verify that $3\sqrt{x} - 5 \geq 0$ and $x \geq 3$ for each.
- Forgetting that $a > 1$ is what makes $f \to \infty$; if $a \leq 1$ there would be no minimum and the picture changes completely.

Beyond STEP

The counting argument rests on two facts from analysis: a continuous function on an interval takes every value between its values at the ends (the intermediate value theorem), and a strictly monotonic function takes each value at most once. Together they turn a sketch into a proof. Abbott's *Understanding Analysis* develops both carefully and shows why continuity is exactly the hypothesis needed.

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